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Spatial Shifts in Italian Extreme Daily Precipitation Revealed by K-means Clustering of Convection-Permitting Climate Projections

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The assessment of extreme precipitation events (EPEs) is crucial due to their significant societal impacts, physical complexity, and critical role in flooding and landslide prediction, as well as their importance for civil protection and water management[1][2][3]. This study investigates extreme daily precipitation patterns across Italy by applying k-means cluster analysis to high-resolution climate projections from the VHR-PRO_IT dataset[4] (2.2 km resolution). Examining both historical (1981-2005) and future periods (2035-2065) under RCP4.5 and RCP8.5 scenarios, the analysis identifies 13 distinct precipitation patterns that highlight the influence of complex orography and geographical features. Results reveal significant seasonal variations and project increased rainfall variability across Italy in future scenarios.

Methodology

The identification of **extreme daily precipitation events** followed a **two-step approach**.

- **First**, candidate events were selected when at least one grid point exceeded its 98th percentile, ensuring local representativeness of extremes.
- **Then**, for those dates, only grid points above their 95th percentile were retained, isolating the most intense areas while preserving spatial coherence.This method balances statistical rigor and spatial detail, producing a robust catalog of extreme precipitation events.

WORKFLOW

Input data (VHR-PRO_IT, 2.2 km)

Pre-processing

Criterion Definition for Identifying Extreme Catalogue Events

1) Selection of dates where at least one point exceeds its **98th** percentile over time.

2) Creation of a new dataset consisting of precipitation values that, for the dates selected in point 1), exceed their **90th** percentile over time.

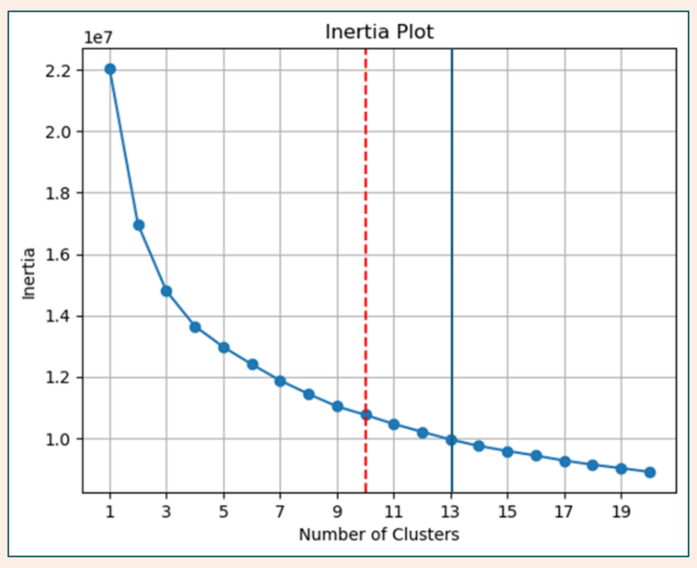
3) Replacing NaN values with 0 in the dataset created in step 2) and **normalization** of the data (or not).

Clustering

K-means

Output

Clusters Matrix

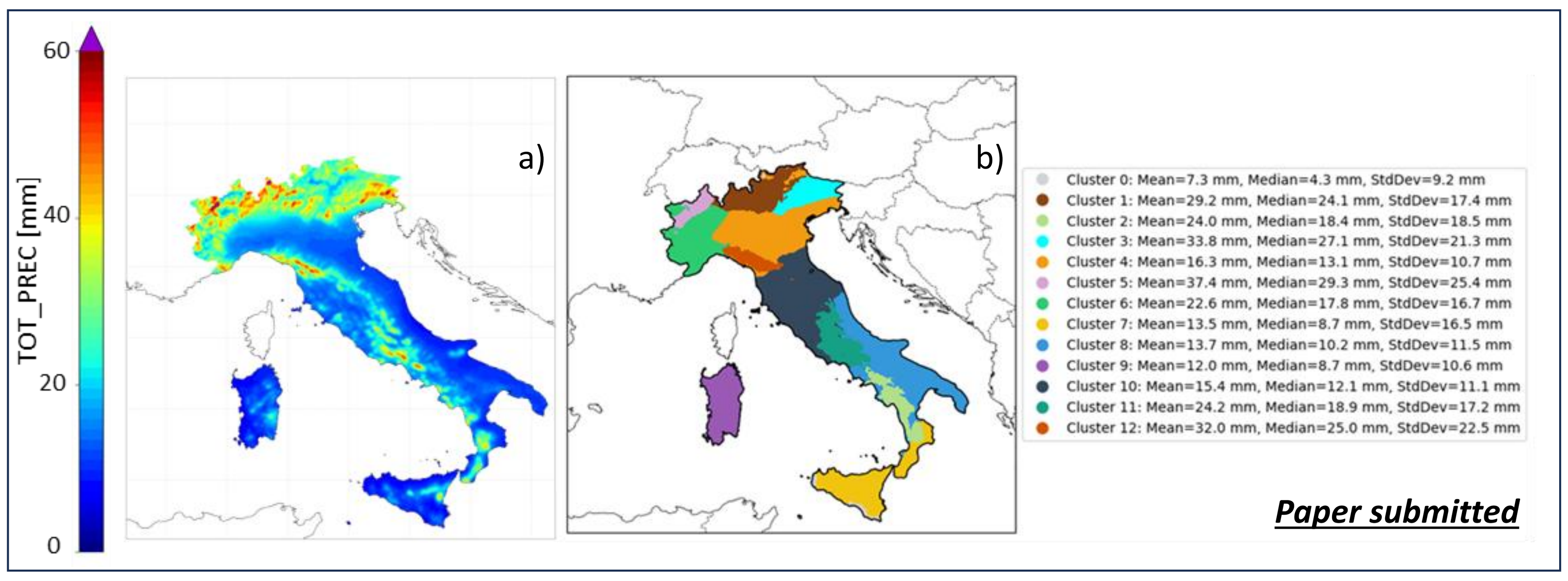


Results

HISTORICAL

Panel (a): Historical extremes (1981–2005):

- Highest values (>30 mm, peaks up to 60 mm) over the Alps and northern Apennines due to orographic lifting.
- Lower values (<15 mm) in the Po Valley and southern inland areas, reflecting continental influence.

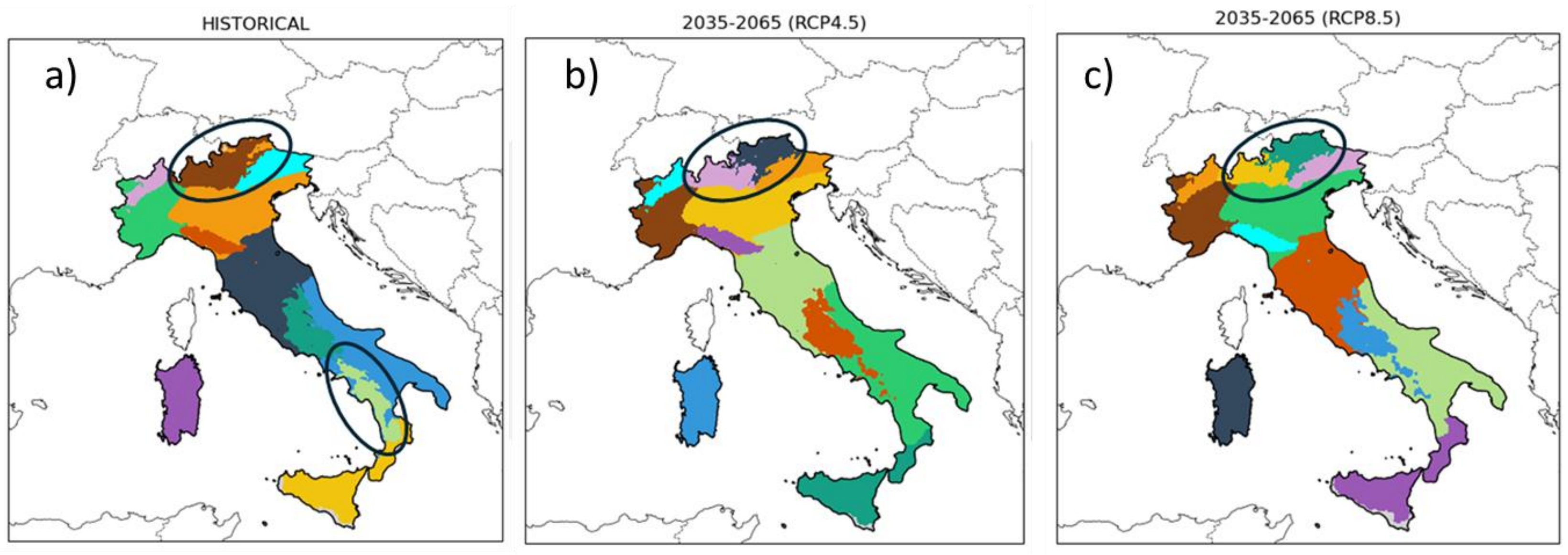


Paper submitted

Panel (b): Clustering results (13 regimes):

- Alpine regions (clusters 1, 3, 5): very intense events (mean up to 37.4 mm; SD > 20 mm).
- Northern Apennines (cluster 12): high precipitation (~32 mm) linked to moist westerly winds.
- Central–southern inland areas and Adriatic side (clusters 8, 10): weaker regimes (~13–15 mm).
- Po Valley (cluster 4): moderate, widespread events (~16 mm) related to large-scale convection and inversions.

FUTURE (RCP.45 & RCP8.5)



Season	Main changes in precipitation	Spatial patterns
Winter	Increase in mean and variability (stronger under RCP8.5).	Piedmont merges with Po Valley; Western Alps show higher values.
Spring	Slight decrease under RCP4.5, small rise under RCP8.5.	More uniform rainfall in Lombardy; stronger precipitation in Apulia, Sicily, and southern Calabria.
Summer	Decrease in extreme rainfall intensity.	Weaker events over Sicily, Sardinia, Calabria, and Tyrrhenian coast; minor changes in Apulia and Campania.
Autumn	Overall similar to historical period.	Western Alps become distinct; stronger increases in Piedmont and Trentino Alps under RCP4.5.

Conclusions

Revealing localized variations in precipitation intensity and shifts in regimes, this approach provides crucial insights for the development of adaptation strategies specific to geographic areas with common precipitation regimes. In operational terms, particularly for civil protection, the clustering methodology can inform the delineation of alert zones.

Next Steps

High-resolution datasets are crucial for accurate classification and vulnerability assessment. Future work will focus on evaluating how using VHR-PRO_IT at hourly instead of daily resolution affects the identification of recurrent precipitation regimes.

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